What the exam expects: 8051 port configuration before use • how LEDs are connected to a port • the AAH/55H alternate-blink program with a delay subroutine. (Context: LED is the first interfacing topic of Module 4 — ADC, DAC, sensors, keyboard and LCD follow.)

1 • STEP 0 — Configure the I/O port before using it

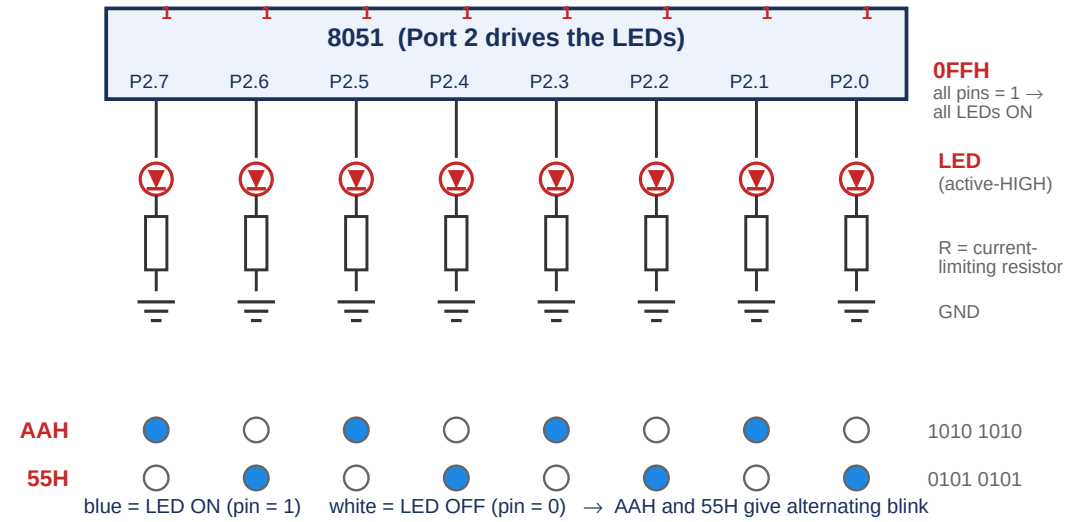
- Always configure the I/O ports **at the beginning** of the ASM program, before any IN/OUT operation.

Goal	Instruction	Why / meaning
Port as INPUT	`MOV P1,#0FFH`	Write 1s to the latch → port acts as input (ready to read)
Port as OUTPUT	`MOV P1,#00H`	Write 0s → port drives the connected device (LEDs)

Memory hook: **FF** = all 1s = **F**ill from outside (input) • `00` = all 0s = **O**utput.

2 • LED interfacing — the concept in 4 lines

- An LED is the simplest **output device**: 8 LEDs are tied to port pins P2.0–P2.7.
- Each pin goes through a **current-limiting resistor** to ground — it limits current and protects the LED and port.
- **Active-HIGH connection:** writing `1` to a pin → LED **ON**; `0` → LED **OFF**.
- One `MOV P2,#data` writes all 8 bits at once → all 8 LEDs change together.
- Slide figure: pins drawn with **1** and labelled `0FFH` → all 1s = **every LED ON**; with AAH/55H only the alternate set lights (patterns under the diagram).



3 • Task: make alternate LEDs blink (ON/OFF)

- Send **AAH (1010 1010)**, delay, then **55H (0101 0101)**, delay … and repeat forever.
- AAH and 55H are **bit-wise complements** → the lit set of LEDs swaps every cycle → blinking pattern.
- The **delay** decides the blink speed — without it the eye sees all LEDs permanently ON.

Read next page → the full annotated program (exactly what the slide asks for).

4 • The program — fully commented (this is THE answer)

```
                        ; ---- MAIN: alternate LEDs ON/OFF on Port 2 ----
ORG 0000H               ; reset address: 8051 starts here after power-on
L1: MOV P2,#0AAH        ; AAh = 1010 1010 → P2.1,3,5,7 HIGH → those LEDs ON
LCALL D1                ; call delay: pattern must stay for a visible time
MOV P2,#55H             ; 55h = 0101 0101 → opposite (even) LEDs ON now
LCALL D1                ; delay again → equal ON/OFF halves = steady blink
SJMP L1                 ; jump back: repeat forever (infinite loop)

                        ; ---- DELAY subroutine (fills the 'write here' gap) ----
D1: MOV R6,#20          ; outer loop counter → sets total delay (blink rate)
D2: MOV R7,#255         ; inner loop counter (255 passes)
D3: DJNZ R7,D3          ; inner loop: decrement R7, loop till 0
DJNZ R6,D2              ; outer loop: repeat inner delay 20 times
RET                     ; return to the instruction after LCALL
END                     ; end of source file
```

5 • One-line explanations the examiner looks for

- **`ORG 0000H`** — on reset the 8051 fetches its first instruction from address 0000H.
- **`AAH / 55H`** — complementary bit patterns (1010 1010 / 0101 0101) → alternate LEDs toggle.
- **`LCALL D1 / RET`** — call the delay subroutine and come back to the next instruction.
- **`SJMP L1`** — short jump back → endless loop so blinking never stops.
- **`DJNZ Rn,label`** — decrement and jump if not zero → builds the time delay (nested loops).

6 • Quick variations (frequently asked)

To do	Change in program	Result
All LEDs ON / OFF	**`MOV P2,#0FFH`** / **`MOV P2,#00H`**	FFH → every pin = 1 (all ON); 00H → all OFF
Blink faster / slower	Change R6 count in D1 (e.g. #05 / #50)	Smaller count = shorter delay = faster blink
Use another port	Replace P2 by P1 (configure it first)	Same program works for P0–P3

30-second revision: Configure port → LEDs on pins via resistors, active-HIGH → loop { P2=AAH, delay, P2=55H, delay } → SJMP repeats forever → delay count sets blink rate.